Sara Bakhtiari

PhD Candidate in Mechanical Engineering | Computational Mechanics | Scientific Computing | Machine Learning & AI

Lexington, KY | Sara.b@uky.edu | LinkedIn | GitHub | Website

Eligible for U.S. internships, subject to university authorization.

PROFESSIONAL SUMMARY

Mechanical Engineering PhD Candidate (GPA 3.91/4.00) at the University of Kentucky specializing in computational mechanics and biomechanics. Develops nonlinear finite-element models coupling passive constitutive behavior, active contraction, circulation, and remodeling. Experienced in Python, FEniCS/DOLFIN, MPI/PETSc, SLURM-based HPC, model validation, scientific post-processing, and machine learning for engineering data.

TECHNICAL SKILLS

Programming & Scientific Computing: Python, MATLAB, NumPy, SciPy, Pandas, FEniCS/DOLFIN, UFL
Machine Learning & AI: scikit-learn, vision-language models, embeddings, regression, cross-validation, feature selection, dimensionality reduction

Computational Mechanics & HPC: Nonlinear FEA, continuum mechanics, constitutive modeling, MPI/mpi4py, PETSc, SLURM, Linux, Singularity

Visualization & Engineering: ParaView, Matplotlib, COMSOL Multiphysics, SolidWorks, HDF5/XDMF

EDUCATION

PhD, Mechanical Engineering (Direct PhD) - University of Kentucky | Spring 2024-Present
Candidate since Apr 2026 | GPA 3.91/4.00 | Advisor: Jonathan F. Wenk, PhD

B.Sc., Mechanical Engineering - Iran University of Science & Technology (IUST) | 2016-2021
GPA 3.55/4.00 | Advisor: Siavash. Kazemirad, PhD

RESEARCH EXPERIENCE

Graduate Research Assistant - Computational Cardiac Mechanics **Spring 2024-Present**
University of Kentucky - Lexington, KY

- Develop and extend MyoFE_H, a coupled finite-element framework integrating ventricular mechanics, passive constitutive laws, half-sarcomere kinetics, active contraction, circulation, growth/remodeling, and fibrosis.
- Evaluate Guccione and Holzapfel-Ogden formulations; quantify effects of material behavior and fiber disarray on stress, deformation, pressure-volume function, efficiency, energetics, and remodeling predictions.
- Run and validate Python/FEniCS simulations using MPI, PETSc, SLURM, and Linux; build workflows for biaxial/shear tests, passive inflation, stress/strain analysis, PV loops, and ParaView post-processing.

Machine Learning & Vision-Language Models for Electrospinning **2026**

- Apply pretrained/fine-tuned VLMs and scikit-learn workflows to video embeddings; evaluate feature variability, learned bottlenecks, regression performance, repeatability, and cross-validation for fiber-diameter prediction.

TEACHING EXPERIENCE

Teaching Assistant - Statics, University of Kentucky **Fall 2026**

- Explain equilibrium, free-body diagrams, force/moment analysis, trusses, frames, and engineering problem-solving methods to undergraduate students.

- Support classroom instruction and student engagement; assist with grading and exam supervision while evaluating solution approaches and communicating technical concepts clearly.

Teaching Assistant - Statics, Iran University of Science and Technology **2021**

- Supported undergraduate engineering-mechanics problem solving and helped students strengthen core statics concepts.

SELECTED PUBLICATIONS & RESEARCH OUTPUTS

Peer-reviewed: Islam, R., Bakhtiari, S., et al. “Advancements in finite element modeling of myocardial infarction: a review.” Biomechanics and Modeling in Mechanobiology, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0

Two first-author manuscripts under review (passive constitutive laws; fiber disarray/efficiency/baroreflex). BMES Annual Meeting poster, San Diego, 2025.